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PRECAMBRIAN PALAEOMAGNETISM IN AUSTRALIA II: BASIC DYKES FROM THE GAWLER BLOCK

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ABSTRACT

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A palaeomagnetic study of fourty Precambrian dykes which intrude the Gawler Block in South Australia, reveals two principal groups of directions after magnetic cleaning. Pole positions computed from dyke-mean V.G.P.s lie at 50.8°E, 61.4°S (A_{95} = 8.6°) for group GA and 86.4°E, 22.8°S (A_{95} = 11.3°) for group GB. Rb—Sr geochronological studies on samples from the two groups indicate that the dykes constituting group GB were intruded at 1700 ± 100 m.y. and the dykes which yield group GA at 1500 ± 200 m.y. Both palaeomagnetic poles coincide with pole positions previously obtained from a study of the South Australian Precambrian haematite ore bodies, thereby providing age constraints on their date of formation.

INTRODUCTION

The Precambrian polar wander curve for Australia has been based on few data (Irving and Green, 1958; Evans, 1968; Chamalaun and Porath, 1968; Porath and Chamalaun, 1968; Facer, 1971; Embleton and Giddings, 1974) gathered from many widely separated localities. A more systematic approach has therefore been adopted in this present study to enable consideration of the results and their interpretation from craton to craton. In view of hypotheses invoking relative motion between blocks of Precambrian crystalline basement, such as proposed by Rod (1966), Arriens (1971) and Crawford and Campbell (1973a), it is necessary to investigate Precambrian rocks from individual crustal nuclei. Hospers and Van Andel (1969) have previously discussed the significance of palaeomagnetism for relating the drift history of individual crustal nuclei. Much the same approach was

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made recently by Piper et al. (1973) during their investigation of the African Precambrian.

In this paper we describe results obtained from a palaeomagnetic study of a swarm of Precambrian dykes which intrude the Gawler Block of South Australia. The maximum age of the basement is considered to be around 1800 m.y. (Compston and Arriens, 1968). We shall also show that the palaeomagnetism of the dykes can assist us in providing closer estimates of the ages of formation of the haematite ore bodies than previously possible (Chamalaun and Porath, 1968). The ore bodies developed from the banded iron formations of the basement (Miles, 1955).

GEOLOGICAL NOTE

The eastern margin of the Gawler Block is bounded by a complex fault structure (the Torrens Hinge Zone — see Fig. 1) which separates it from the younger Adelaide Geosyncline. A sequence in excess of 15 km of Upper Precambrian (Adelaidean System) to Cambrian sediments accumulated in the geosyncline. Deformation took place during the Delamerian Orogeny

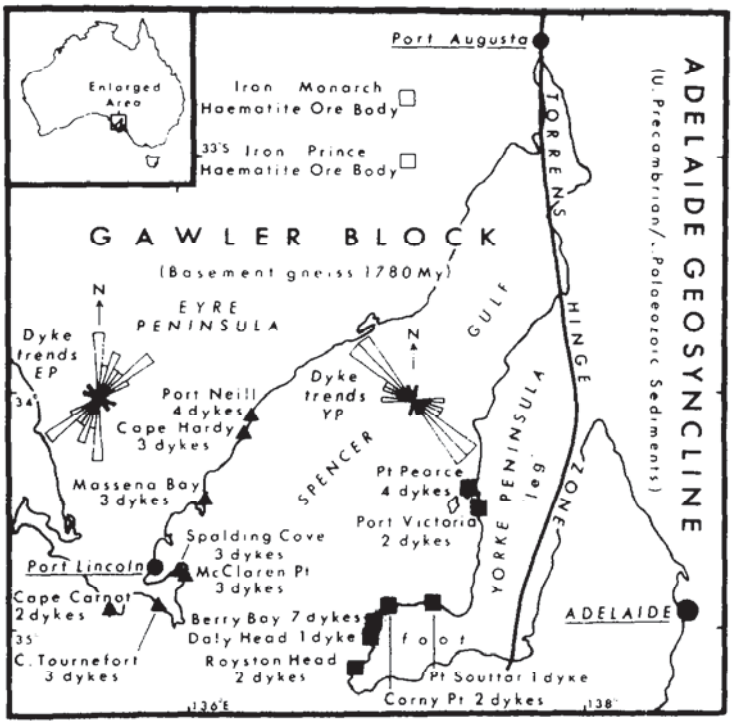


Fig. 1. The southeastern corner of the Gawler Block showing the sampling localities of the dykes and their trends (inset compass-rose diagrams) and the location of the haematite ore bodies studied by Chamalaun and Porath (1968).

(Thomson, 1969) in the Late Cambrian and Ordovician. Less complete sequences occur in isolated outcrops along the eastern part of the Gawler Block. They are correlated with sediments in the Adelaide Geosyncline proper but are mainly flat-lying (Johns, 1961; Sprigg, 1961; Crawford, 1965). We therefore regard that region of Precambrian to have been little affected by tectonic disturbances and that the attitudes of the dykes have not altered significantly since intrusion. The dykes outcrop along the facing coastlines of the Spencer Gulf on southern Eyre Peninsula and Yorke Peninsula.

Southern Eyre Peninsula

The foundation of the area is a broad band of Proterozoic metasediments which have been tightly folded into a series of north-trending synclines and anticlines (Johns, 1961). The sequence has been divided into an older gneissose part (the Flinders Group) and a younger schistose part (the Hutchison Group). The 1800 m.y. date was obtained from granulites of the Flinders Group. The basic dykes generally occur as concordant bodies in the gneiss. Their field occurrence is either that of elongated, lenticular, dark bands often penetrated by veinlets of country rock, or as intrusive, massive bodies with chilled contacts and without visible country rock contamination. Tilley (1921) made an extensive study of the dykes and divided them into metadolerites (granulite grade rocks) and various types of amphibolites. Johns (1961) has recorded dykes with primary doleritic textures. The compass-rose in Fig. 1 shows the distribution of the dyke-trends measured in the field during sampling. No evidence of dykes cross-cutting themselves was observed. The dykes best outcrop along the coast where active erosion provides fresh material. On southern Eyre Peninsula at localities in Fig. 1, 21 dykes were sampled.

An haematite quartzite near the base of the Hutchison Group is considered by Whitten (1966) to be equivalent to the haematite quartzites of the Middleback Group (Miles, 1955) which outcrop in the Middleback Ranges in the northern part of Eyre Peninsula. In those ranges, the quartzites are host to the high-grade haematite iron ore bodies studied by Chamalaun and Porath (1968). Figure 1 shows the locations of two of the ore bodies, Iron Monarch and Iron Prince, for which palaeomagnetic data are available.

Yorke Peninsula

Regional mapping by Crawford (1965) has shown that the basement consists of a strongly folded sequence of gneissic metasediments. Their relationship to those of Eyre Peninsula, however, is obscure. Radiometric ages for them are not available, but dated intrusives suggest a minimum age of about 1500 m.y. (Compston et al., 1966; Thomson, 1972). The dykes,

described by Crawford (1965) as amphibolites, are generally concordant with the foliation in the metasediments and are nowhere found to intrude the overlying thin developments of Adelaidean (Upper Precambrian) sediments. Since over 90% of the peninsula is mantled with Quaternary deposits, outcrop of the basement and dykes is confined to isolated localities along the coast (see Fig. 1). Nineteen dykes were sampled in the intertidal zone. The general northwesterly strike of the six dykes sampled on the 'leg' of the peninsula paralleled the strike measured in the dykes collected from the 'foot' of the peninsula.

SAMPLING AND MEASUREMENT DETAILS

Only massive dykes, with well-defined margins and without visible country rock contamination were sampled. Samples were collected as blocks and oriented with both the sun and magnetic compass techniques. At least three samples were collected from each dyke, and where possible were distributed across the width, yielding a total of 131 samples. In the laboratory they were cored to provide three or four cylindrical specimens, 28 mm × 28 mm. Measurements of intensity and direction of magnetic remanence were made with a P.A.R. model SM1 spinner magnetometer. A statistical analysis of the directions of magnetization was made using the method devised by Fisher (1953). Randomness of a set of directions was tested by applying the criteria of Watson (1956) and Vincenz and Bruckshaw (1960). Specimens were cleaned with the alternating magnetic field technique using equipment similar to that described by McElhinny (1966).

DESCRIPTION OF RESULTS

Twenty dykes yielded mean directions of natural remanent magnetization (NRM) considered non-random. Broadly, they fall into two principal groups with opposite polarities, though not antiparallel — see Fig. 2. NRM directions in the remaining twenty dykes were shown to be random at the 95% probability level. The scatter was due to streaking towards the present field axis or in the case of the more weakly magnetized dykes to an apparent lack of any preferred orientation.

An analysis of the stability of the magnetic remanence was undertaken by subjecting 71 pilot specimens from 35 dykes to stepwise AF demagnetization until the coercive force of the stable remanence was exceeded (i.e. until the onset of apparently erratic response of direction and intensity of remanence). Giddings and McElhinny (in press) have developed a stability index which properly describes the stability of magnetic remanence as a function of vector differences between successive increments of the applied alternating field (similar to that devised by Briden, 1972) and the coercive force spectrum over which end-point behaviour is considered to have been attained. The range of stability indices calculated in the present study are

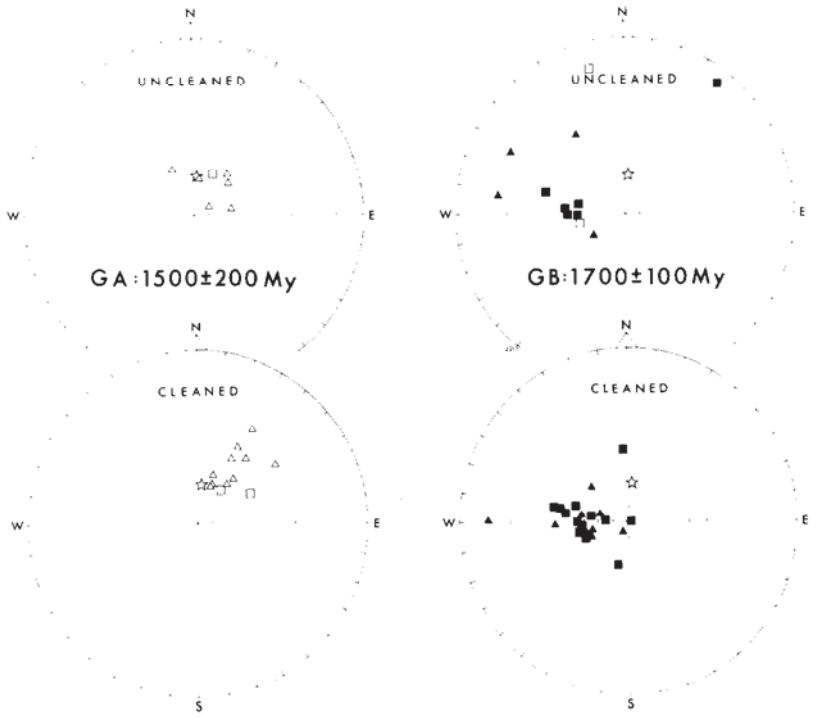


Fig. 2. Dyke-mean directions of magnetization of the South Australian dykes before and after magnetic cleaning. The cleaned directions fall into two distinct groups, labelled GA and GB. Directions, plotted on Wulff equal-angle projections, are denoted by solid (open) symbols if they are downward (upward) pointing. Squares (triangles) refer to Yorke Peninsula (Eyre Peninsula) dykes. The star represents the direction of the earth's present magnetic field in the sampling area.

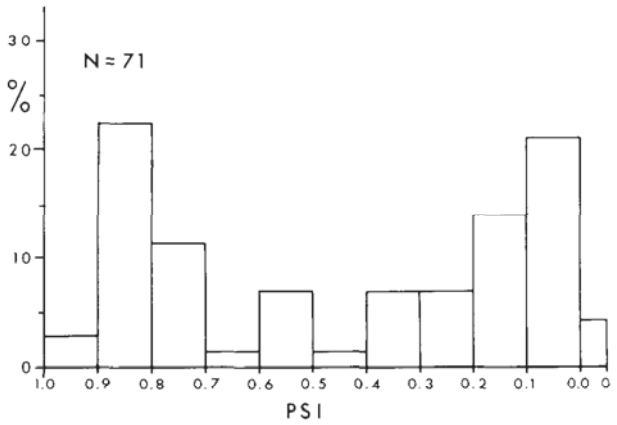


Fig. 3. The stability characteristics of the dykes expressed in terms of the palaeomagnetic stability index (PSI) defined by Giddings and McElhinny (in press). *N* represents the number of pilot specimens used in the analysis. The spectrum of possible PSI values ($1.0 \leqslant PSI \leqslant 0$) is characterized by two peaks. The peak at values of $PSI \geqslant 0.80$ corresponds to extremely stable remanence whose origin must lie in single-domain or pseudo-single domain grains; the peak at low values of PSI ($\leqslant 0.11$) corresponds to partially stable or unstable remanence.

TABLE I

Summary of dyke-mean palaeomagnetic data after magnetic cleaning

Locality	Dyke No.	N(Sa)*	R	Dec. (°)	Inc. (°)	Group
Eyre Peninsula						
Port Neill	1	9(3)	8.96	17.5	−63.6	GA
136.3°E 34.1°S	2	3(2)	2.97	37.0	−39.4	GA
	3	9(3)	8.58	31.3	−25.6	GA
	4	7(3)	6.60	283.6	70.2	GB
Cape Hardy	5	9(3)	8.31	20.3	−62.9	GA
136.3°E 34.2°S	6	8(3)	7.75	28.6	−36.4	GA
	7*1	—	—	—	—	—
Marsena Bay	8	9(3)	8.991	257.9	58.7	GB
136.1°E 34.5°S	9	8(3)	7.59	267.1	43.4	GB
	10	6(2)	5.97	210.0	81.9	GB
Spalding Cove	11	9(3)	8.96	248.5	62.8	GB
136°E 34.8°S	12	7(3)	6.56	255.3	65.2	GB
	13	9(3)	8.57	277.2	58.3	GB
McClaren Point	14	9(3)	7.85	312.0	56.7	GB
136°E 34.8°S	15*1	—	—	—	—	—
	16	8(3)	7.13	18.1	−57.3	GA
Cape Tournefort	17	9(3)	8.94	36.6	−58.2	GA
135.9°E 34.9°S	18	9(3)	8.92	27.4	−44.0	GA
	19	9(3)	8.43	38.8	−53.3	GA
Cape Carnot	20	5(2)	4.59	53.6	−30.4	GA
135.6°E 34.9°S	21	9(3)	8.30	271.2	11.7	GB
Yorke Peninsula						
Souttar Point	22	15(5)	12.77	356.5	45.2	GB
137.3°E 34.9°S						
Corny Point	23*2	2(1)	1.96	265.0	45.7	GB
137°E 34.9°S	24*1	—	—	—	—	—
Berry Bay	25	9(3)	8.996	257.3	57.2	GB
137°E 34.9°S	26*2	3(1)	2.97	285.9	68.5	GB
	27	6(2)	5.81	59.8	−51.2	GA
	28	7(3)	6.21	194.7	60.9	GB
	29	9(3)	8.98	268.8	56.5	GB
	30	9(3)	8.99	265.6	58.8	GB
	31	5(2)	4.79	269.7	75.1	GB
Daly Head	32*2	3(1)	2.81	171.7	77.6	GB
136.9°E 35°S						
Royston Head	33	9(3)	8.98	277.6	65.3	GB
136.9°E 35.2°S	34	9(3)	8.992	281.3	42.6	GB
Port Victoria	35	11(4)	10.39	35.4	−63.3	GA
137.5°E 34.5°S	36	6(2)	5.59	103.9	88.7	GB
Point Pearce	37	9(3)	8.72	285.6	54.3	GB
137.4°E 34.5°S	38	9(3)	8.66	249.4	59.7	GB
	39	12(4)	11.35	277.6	49.0	GB
	40	12(4)	11.72	280.8	45.6	GB

* N(Sa) = specimens (samples) per dyke and R = resultant of N unit vectors.

*1 No specimens from these three dykes contained magnetic remanence with stable end-point behaviour. SI values were less than 0.1.

*2 Specimens from only one sample yielded SI values greater than 0.1. Results from these three dykes were omitted from the palaeomagnetic pole calculations.

TABLE II

Group-mean directions and palaeomagnetic pole positions for the Precambrian dykes from South Australia.

Group	Age	N	R	Dec. (°)	Inc. (°)	α_{95} (°)	Pole position		A_{95} (°)
							lat. (°)	long. (°)	
GA	1500 ± 200 m.y.	12	11.589	34.7	−49.5	8.5	61.4S	50.8E	8.6
GB	1700 ± 100 m.y.	22	20.510	272.5	+61.1	8.5	22.8S	86.4E	11.3

shown in the histogram in Fig. 3. A complete spectrum of magnetic stability is represented by the collection. The histogram peaks at two points, one at PSI values less than 0.10 which reflect only partially or totally unstable magnetic remanence and the other at PSI values greater than 0.80 which correspond to extremely stable remanence. For rocks in the latter category the coercive force of the remanence exceeds 80 mT. This would indicate that the magnetic particles behave in a single domain or pseudo-single domain fashion since the carrier of the remanence is titanomagnetite (Evans et al., 1968). However, PSI values greater than 0.11 indicate that over a particular coercive force spectrum, a well defined end-point is reached. Suitable cleaning fields for the remainder of the material could then be chosen. Table I lists the dyke-mean directions of magnetization after magnetic cleaning; unit weight was assigned to specimens. This procedure was preferred since ten dykes exhibited a marked within-dyke variation of magnetic stability so that sample-means based on only one or two specimens would be unfairly weighted against samples based on three or four specimens. To remain consistent throughout, the specimen was taken as the unit of observation. Stable magnetic remanence was isolated in 34 dykes. Their directions fall in two distinct groups labelled GA and GB in Fig. 2. Table II lists the group-mean directions, the pole positions and estimates of their precision.

DISCUSSION

There is little doubt that directions of magnetization which constitute group GB refer to the primary component of magnetization. However, group GA is magnetized close to the present direction of the earth's magnetic field. Nevertheless, we also regard its mean direction as equally valid but probably imposed during a separate episode of dyke intrusion. Selective remagnetization is unlikely and a regional thermal event is discounted since the two groups of directions were obtained from dykes sampled in nearby localities. Furthermore, magnetic remanence of each group revealed coercive forces in excess of 80 mT. This indication of single-domain be-

haviour is very important in view of the possibility of low-temperature oxidation of single-domain magnetites (Johnson and Merrill, in press) which allows a grain to oxidize yet still record its initial direction of magnetic remanence, irrespective of the ambient field during alteration. The possibility of significant systematic deflection of the primary components of remanence, as one of the by-products of metamorphism, is discounted. Within each group, dykes which span a spectrum of metamorphic grade display consistent directions of magnetization. Irving and Park (1973) have previously demonstrated that metamorphic foliation introduces a random component of magnetization.

The positions of the palaeomagnetic poles were computed assigning unit weight to dyke-mean directions (Table II). The poles, plotted in Fig. 4, are separated by about 45°. The three poles obtained by Chamalaun and Porath (1968) from their study of the Iron Monarch and Iron Prince ore bodies, mentioned in the Introduction, are also plotted in Fig. 4 for comparison. The absolute age of the ore bodies remained largely unknown other than they were Precambrian and the palaeomagnetic data suggested that they were formed in three principal phases. The two poles yielded by the Iron Monarch ore body agree with the poles yielded by the Precambrian dykes and the pole obtained from Iron Prince lies between the

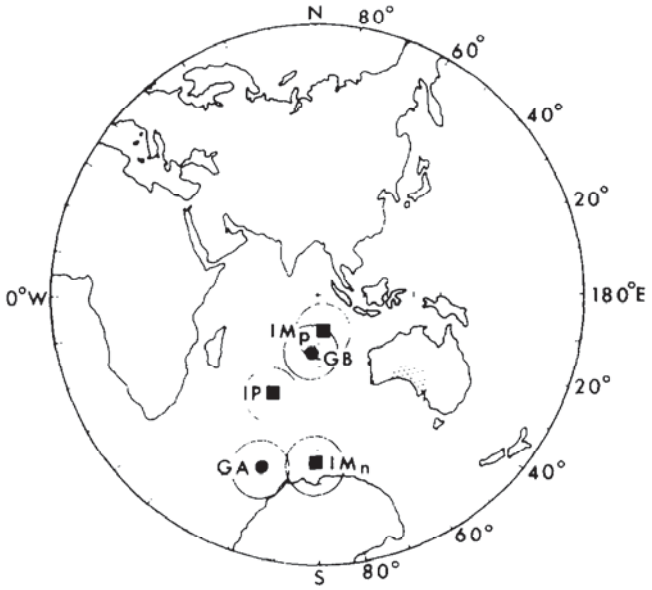


Fig. 4. Pole positions and associated circles of confidence at the 95% level for the dykes (GA and GB) compared with the poles obtained by Chamalaun and Porath (1968) from the haematite ore bodies of the Middleback Ranges, Eyre Peninsula. IM_n and IM_p refer to the two poles obtained from the Iron Monarch ore body and IP refers to the pole from the Iron Prince ore body. The close agreement of the ore body poles with those of the dykes permits narrow constraints to be placed on the ages of ore formation.

two groups. Compston and Crawford (pers. comm., 1975) have investigated the Rb—Sr geochronology of the dykes. Group GB dykes yield an age of 1700 ± 100 m.y. and group GA dykes are younger at 1500 ± 200 m.y. Narrow age constraints may therefore be inferred for the periods of ore formation in the South Australian Middleback Ranges. This investigation is part of a larger corroborative research programme to study the palaeomagnetism of basic intrusives located in the major Precambrian crustal blocks which constitute the main Australian platform. Results from the Yilgarn Block have been described by Giddings (in press) and results from the Kimberley and Mount Isa Blocks have been reported by McElhinny and Evans (in press) and Duff and Embleton (in press), respectively.

The palaeomagnetic results from the southeastern region of the Gawler Block also have an important bearing on questions concerning the structural history of that region. Crawford and Campbell (1973a) proposed a model involving the rotation of Yorke Peninsula with respect to Eyre Peninsula and that the 'foot' of Yorke Peninsula was snapped through 90° relative to the 'leg' of the peninsula in the Early Ordovician. Group GB comprises directions of remanent magnetization measured in dykes from southern Eyre Peninsula and the 'foot' and 'leg' of Yorke Peninsula. The case made out by Giddings and Embleton (1974) based on the data presented here refutes relative rotation between those regions and was acknowledged by Crawford and Campbell (1973b).

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